

MATH 1342 – Elementary Statistics – Week 8: Percentile Intervals and Introducing Hypothesis Tests

Oct 13 – 15 | 3.4 Bootstrap Confidence Intervals Using Percentiles • 4.1 Introducing Hypothesis Tests | Fall 2026 · Section 036 · TTh 6:00–7:20 PM (DIL)

Module Objectives	Instructional Materials	Learning Activities	Assessments	Tools
1. Determine which percentiles of a bootstrap distribution bound a 90%, 95%, or 99% confidence interval, and construct the interval with StatKey. (CLO 3, CLO 6) 2. Predict how changing the confidence level or the sample size affects the width of an interval. (CLO 6) 3. Judge whether a bootstrap confidence interval is trustworthy from the shape of the bootstrap distribution and whether the sample is representative. (CLO 6)	e-Text: 3.4 Bootstrap Confidence Intervals Using Percentiles Interactive Lecture: 3.4 Bootstrap Confidence Intervals Using Percentiles Slides: 3.4 Bootstrap Confidence Intervals Using Percentiles	Participation: Percentile Confidence Intervals – StatKey: 90/95/99% intervals for the same data; compare SE and percentile methods	Online Homework: 3.4 Bootstrap Confidence Intervals Using Percentiles (due Sun Oct 18, 11:59 PM)	WileyPLUS StatKey Lecture Slides (course website)
4. Identify the parameter of interest and write null and alternative hypotheses from a research question. (CLO 8) 5. Classify a test as right-tailed, left-tailed, or two-tailed, and explain the logic of assuming H_0 and asking whether the data are surprising. (CLO 8)	e-Text: 4.1 Introducing Hypothesis Tests Interactive Lecture: 4.1 Introducing Hypothesis Tests Slides: 4.1 Introducing Hypothesis Tests	Participation: Writing Hypotheses – Zener-card ESP practice (26 of 50): define the parameter, state H_0 and H_a, choose the tail Think about it: is 15 of 16 correct (dolphin study) convincing?	Online Homework: 4.1 Introducing Hypothesis Tests (due Sun Oct 18, 11:59 PM)	WileyPLUS StatKey Lecture Slides (course website) Zoom (live class)
<i>Course resources and assessments of earlier weeks (not aligned to this week's objectives)</i>			Take-Home Quiz 4 (due Tue Oct 13, 11:59 PM) – covers P.1, 3.1–3.3	Assignment Tool in Blackboard

CLO 3 Compute and interpret empirical and theoretical probabilities using the rules of probabilities and combinatorics. **CLO 6** Describe and compute confidence intervals. **CLO 8** Perform hypothesis testing using statistical methods.